



CIS 635 Knowledge Discovery & Data Mining

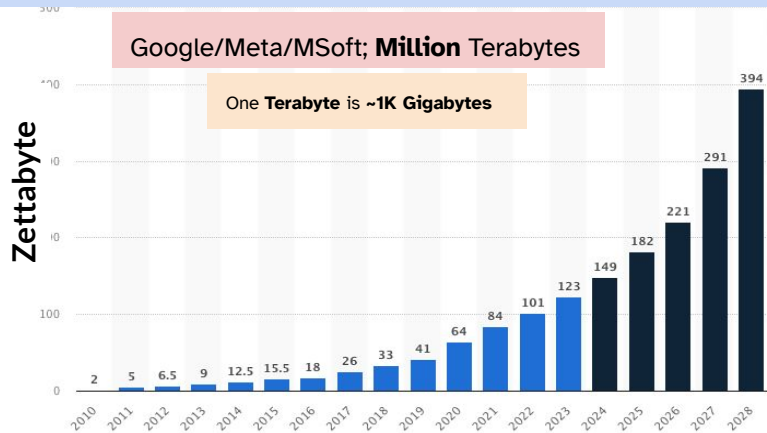
Data Growth and Why Data Science is Critical Today?

Data Growth and the Trend



Data Growth and the Trend

One **Zettabyte** is approximately equal 1 billion Terabytes

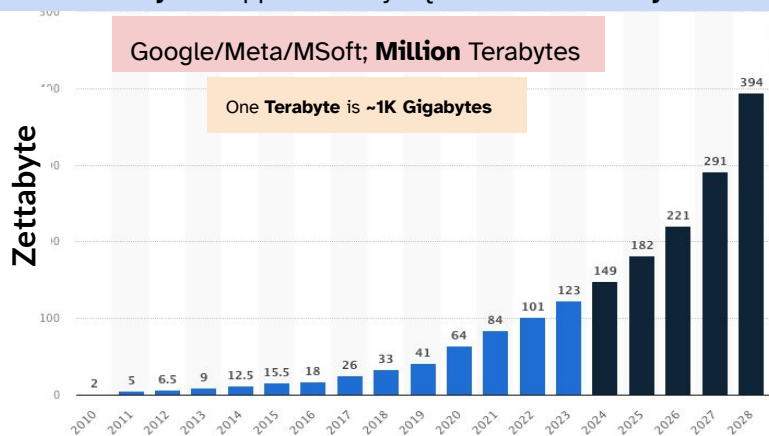


[Volume of data/information created, captured, copied, and consumed worldwide from 2010 to 2023, with forecasts from 2024 to 2028 \(in Zettabytes\)](#)

- **Exponential growth**
 - *Before 2010 collectively relatively close to Zero Zetabyte*
- **Mainly due to**
 - IoT sensors
 - Social media
 -
- **Forecast might be underestimated as we expect more due to the recent Generative AI/LLM releases (ChatGPT for an example)**

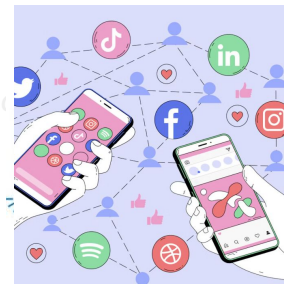
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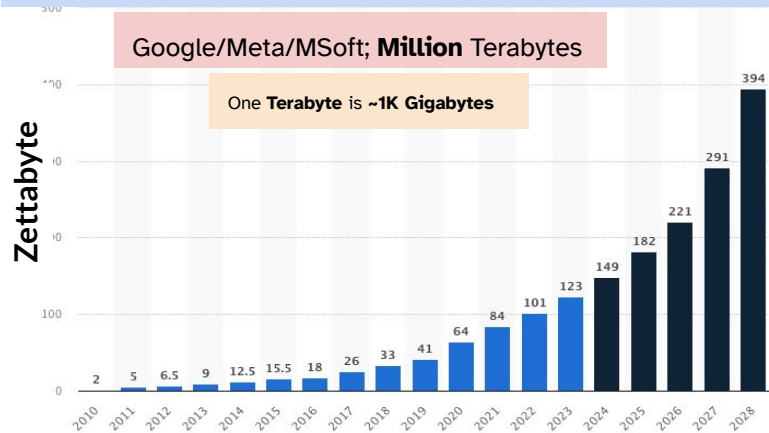
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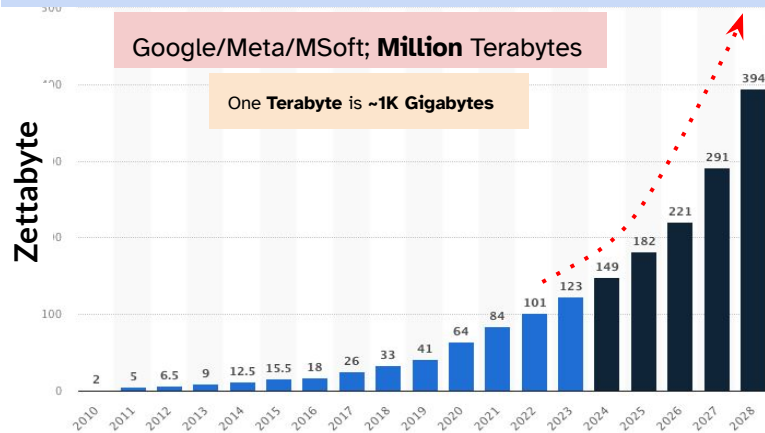
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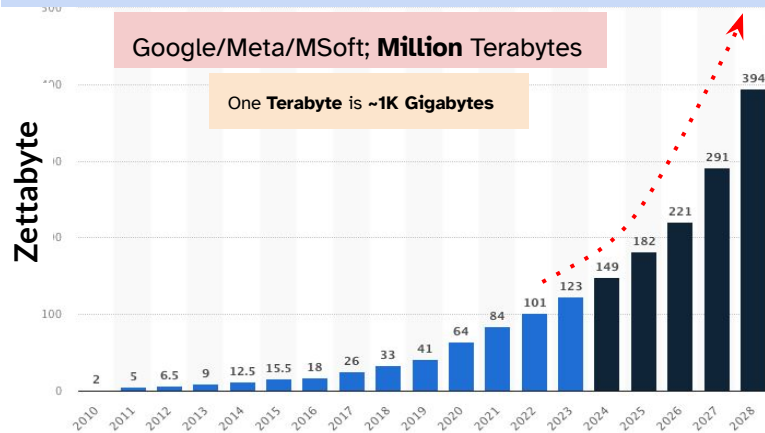
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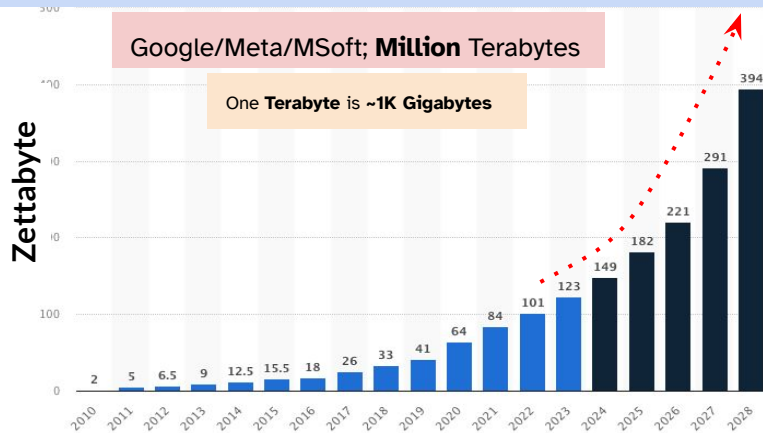


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- **Data Science has emerged as an important discipline**
- We need to tackle new challenges, and many more are yet to come
- We will be learn about modern day **pioneers** in our class moving forward
- However, most the basic principles relates to basics of Math, Statistics and Probability theories. **Tribute** to

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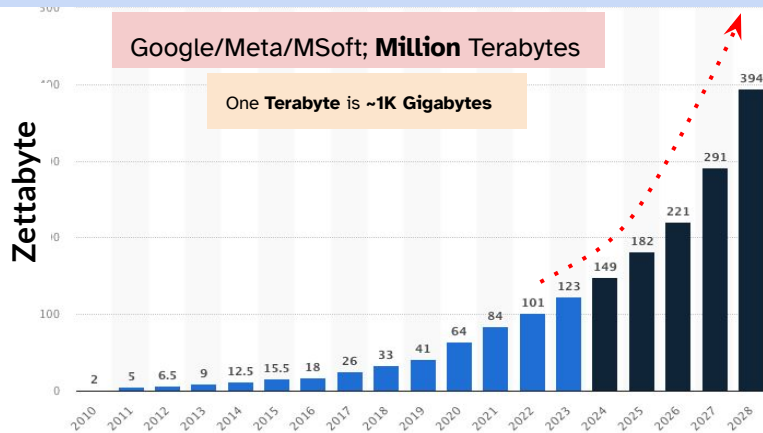


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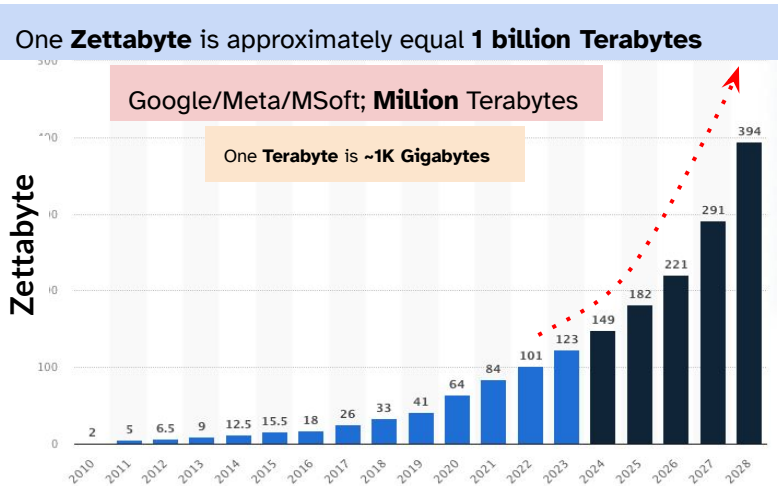
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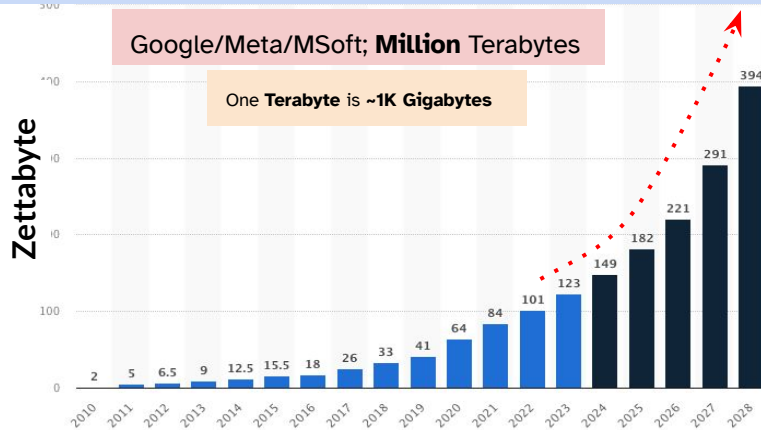


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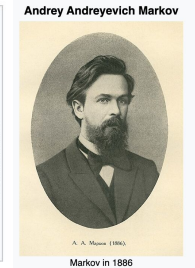
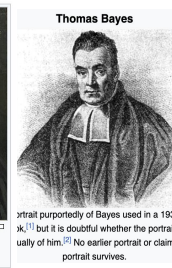
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